EnvelopeTC: Hierarchical Intra-Path Learning for Adaptive Traffic Control

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***Abstract*—Software defined networks offer global visibility, yet centralized control loops are too slow for transient congestion and bursty traffic dynamics. Existing learned traffic control schemes often rely on offline training, making them fragile under distribution shifts. We present EnvelopeTC, a hierarchical SDN traffic control framework that enables local online adaptation under centralized policy control. Its key abstraction is a policy envelope: the controller compiles network wide intent into bounded per path action spaces, while edge agents learn and execute metering, queueing, and rerouting decisions only within those bounds. Policy envelopes also make local actions auditable and reversible when they affect shared bottlenecks. Evaluation on a 1,024 host software SDN testbed shows that EnvelopeTC improves average core link utilization by 35.5% over Static ECMP and 18.3% over Centralized TE. It reduces elephant flow P99 FCT by 34.3% over end host congestion control, lowers SLA violations from 18.2% to 6.8%, and uses less than 2% CPU and 12 MB memory per edge agent.**
The source code is available in an anonymized repository at https://anonymous.4open.science/r/JCC2026-EnvelopeTC/.

***Index Terms*—software-defined networking, online learning, traffic engineering, congestion control, intra-path learning, Hoeffding Adaptive Tree**

I. Introduction

Software-defined networking has become the dominant paradigm for operating large-scale data-center and wide-area networks. By centralizing control in a programmable controller, SDN enables fine-grained traffic-engineering policies, tenant isolation, and network-wide topology responses from a single vantage point [1], [2].

This global visibility, however, creates a fundamental tension with fast traffic control. Congestion events, elephant-flow bursts, and link microbursts evolve on timescales of tens of milliseconds, comparable to a single RTT. Routing per-RTT control decisions through a centralized controller incurs at least one controller round-trip plus processing delay, making purely centralized fast-loop control impractical at scale [3].

The networking community has responded with two broad strategies. *Static policy* programs switch pipelines with pre-computed ECMP weights, meter rules, or queue assignments; it is fast but inflexible, as it cannot adapt to dynamic traffic matrices or link-quality changes. *Offline learning* trains a neural network or RL agent on historical traces and deploys the resulting policy to the controller or data plane; this approach adapts within the training distribution but fails to generalize when real traffic deviates from training conditions, a well-documented limitation of inter-flow learning [4].

This paper presents EnvelopeTC, a hierarchical SDN traffic-control architecture based on *intra-path learning*: edge agents learn and execute fast local actions while the controller manages global intent, action bounds, and coordination. The contribution is not a new learning model; it is a control abstraction that makes locality-based online learning safe and composable for SDN actions whose effects extend beyond the local learning scope.

EnvelopeTC makes three contributions:

- **Intra-path learning.** We scope each policy cache to a path, bottleneck, or tenant aggregate; training and execution are confined to that aggregate only.
- **Policy envelopes.** The controller computes rate bounds, reroute permissions, and utility weights; agents enforce these envelopes before issuing any switch action.
- **Safe multi-agent coordination.** Bottleneck alerts, action logs, and primary-agent arbitration serialize conflicts. We implement EnvelopeTC with Ryu, Open vSwitch, and gRPC, and evaluate it against nine baselines on a 1,024-host testbed.

II. Background and Motivation

A. SDN Traffic Engineering Today

Modern SDN deployments address traffic engineering at two timescales: the *slow timescale* (seconds to minutes), where the controller installs or updates forwarding rules in response to topology events and traffic-matrix shifts; and the *fast timescale* (sub-second), where individual flows encounter congestion requiring millisecond-level rate adjustment, rerouting, or queue management.

Existing mechanisms handle fast-timescale control poorly. Static OpenFlow meter rules apply fixed rate caps regardless of load; ECMP uses fixed weights that cannot react to transient hot spots; and threshold-based rerouting triggers only above a static utilization threshold, causing oscillation under dynamic traffic.

B. The Generalization Problem in SDN

Several works have proposed replacing hand-crafted SDN policies with learned ones. Offline reinforcement learning [5] trains an agent on simulated or historical traces and deploys it to the controller. These approaches share a common limitation: the fixed policy degrades when real conditions differ from the training distribution (the *inter-flow generalization* problem [6]).

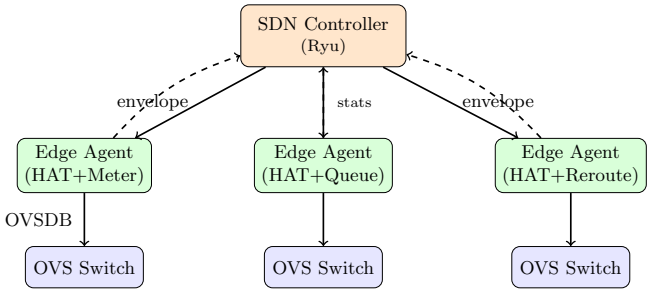


Fig. 1. EnvelopeTC two-plane architecture. The SDN controller manages global state and distributes policy envelopes; edge agents run intra-path learning and execute fast SDN actions.

Deploying an Aurora-architecture agent [4] under shifted traffic matrices or link-failure hot spots degrades utility by up to 40%, consistent with this limitation.

C. From Intra-Flow to Intra-Path Learning

PolicyCache [6] proposes *intra-flow learning*: training and execution are confined to a single flow, avoiding cross-flow generalization. EnvelopeTC reuses this locality principle and the same non-parametric HAT family, lifting the scope to an SDN *path aggregate*: an active path, bottleneck, tenant, or service class observed and controlled by one edge agent. The learned policy applies only to that aggregate and is invalidated when path conditions change. This lift introduces systems challenges absent in single-flow CC: rerouting can displace congestion to another link, meter updates can affect co-located tenants, and two agents on the same bottleneck can oscillate. EnvelopeTC addresses these through controller-compiled policy envelopes, shared-resource monitoring, and conflict arbitration.

III. EnvelopeTC Architecture

EnvelopeTC consists of two control planes operating at different timescales and granularities (Figure 1).

A. Central SDN Controller: Slow Global Control

The SDN controller operates at the *slow timescale* (hundreds of milliseconds to seconds), maintaining the topology graph and backup paths, aggregating per-link telemetry from sFlow/INT streams, computing policy envelopes for active aggregates, detecting and arbitrating conflicting agent actions at shared bottlenecks, and monitoring agent behavior with override and rollback capability.

Envelope computation. The controller computes envelopes from measured demand, topology, tenant policy, and service templates. For each congested link b with capacity C_b and active aggregates A_b, it reserves tenant floors m_i and headroom $\epsilon_q C_b$; the remaining budget $H_b = (1-\epsilon_q)C_b - \sum_{j\in A_b} m_j$ is allocated by weighted water filling:

$$r_{\min,i} = m_i,$$
$$r_{\max,i} = \min\Big\{\hat{d}_i(1+\rho),\ R_i^{\text{tenant}},\ m_i + H_b\frac{\phi_i}{\sum_{j\in A_b}\phi_j}\Big\}.$$

For a path crossing multiple constrained links, the agent receives the tightest per-link bound, making fairness and tenant isolation controller-enforced: an agent may learn how to use its allocation but cannot exceed tenant ceilings or reroute outside permitted paths.

Utility weights. The utility vector $\mathbf{w}$ is initialized from service-class templates and adjusted at each refresh: sustained SLA violations increase w_{lat} and w_{sla}; spare capacity with satisfied SLAs increases w_{thr}. Weights are normalized and clipped to operator ranges, so utility adaptation changes the local objective without granting new action authority.

A policy envelope is the contract that constrains exploration, clips model predictions before execution, provides a versioned audit object, and enables safe fallback when stale. The controller does *not* make per-RTT decisions; it bounds which per-interval decisions are legal.

B. Edge Agent: Fast Local Online Learning

Each *edge agent* is deployed at a ToR switch CPU, vSwitch, P4 control plane, or SmartNIC. It maintains one *policy cache* per monitored path or traffic aggregate, implemented as a Hoeffding Adaptive Tree (HAT) as in PolicyCache [6].

The agent operates in two modes. In **backup exploration mode**, when the cache is untrained or underperforming ($p < p_{\text{th}}$), it probes envelope-valid candidate actions on canary traffic and uses the higher-utility action as the empirical label. In **model execution mode**, once $p \geq p_{\text{th}}$, the HAT predicts an action, the enforcer clips it to the envelope, reroutes are shadow-checked, and the committed action is logged for audit and rollback. If the envelope becomes stale, agents disable new reroutes and keep only meter/queue actions inside the last valid bounds. ADWIN [7] triggers a return to exploration when path conditions shift.

C. Controller–Agent Interface

The interface is lightweight and asynchronous: the controller pushes policy envelopes, bottleneck alerts, and overrides, while agents return telemetry, model state, and action logs. Envelopes carry a version and staleness bound; agents apply updates on the next action cycle without requiring flow-level rule refreshes. This is what distinguishes EnvelopeTC from a plain intra-flow learner on an SDN switch: local learning is permitted only inside a controller-compiled action set.

IV. Learning Formulation and Analysis

A. State

The state vector observed by an edge agent at each measurement interval t is:

$$\mathbf{s}_t = [\text{util}_t,\ q_t,\ \text{loss}_t,\ \text{ecn}_t,\ \text{thr}_t,\ \text{delay}_t,\ \Delta\text{util}_{t-1},\ \Delta\text{thr}_{t-1},\ \Delta\text{delay}_{t-1},\ a_{t-1},\ a_{t-2}]$$

where $\Delta(\cdot)$ denotes the one-interval relative change. Relative-change features capture temporal dynamics without requiring cross-path normalization, consistent with the intra-path learning philosophy.

B. Action

EnvelopeTC controls SDN-level actions rather than TCP cwnd. The action space is discrete and path-specific; at each interval the agent selects one action per dimension: **meter-rate adjustment** (increase or decrease by step α, clipped to $[r_{\min}, r_{\max}]$); **queue assignment** (promote or demote the flow class by one priority level); and **elephant-flow rerouting** (binary trigger/release for flows exceeding 10 MB in the measurement window). ECN marks are used as telemetry features; the prototype does not tune ECN marking thresholds.

C. Utility Function

$$u_t = w_{\text{thr}}\,\Delta\text{thr}_t - w_{\text{lat}}\,\Delta\text{delay}_t - w_{\text{loss}}\,\text{loss}_t - w_{\text{sla}}\,v_t - w_{\text{act}}\,|a_t - a_{t-1}| \quad (1)$$

where v_t is the SLA violation indicator. The controller distributes a utility vector $\mathbf{w} = (w_{\text{thr}}, w_{\text{lat}}, w_{\text{loss}}, w_{\text{sla}}, w_{\text{act}})$ as part of the policy envelope, using the service-class templates and bounded adjustment rules described in Section III.

D. Empirical Labels

EnvelopeTC does not assume access to a global optimal action. During backup exploration, the agent obtains an *empirical label* $a^\dagger(\mathbf{s})$ by comparing two envelope-valid candidate actions over adjacent measurement intervals and selecting the one with higher observed utility. Thus $a^\dagger$ is a local, path-specific target, not a network-wide optimum, and remains valid only while the local distribution, envelope, and neighboring agents' actions are sufficiently stable over the exploration window.

E. Envelope Refresh and Guarantees

At each slow-timescale refresh, the controller initializes each aggregate's floor and cap from tenant policy, tightens $r_{\max}$ on congested links by weighted water filling, sets the reroute permission bit from residual capacity and cooldown state, and pushes the versioned envelope $\langle r_{\min}, r_{\max}, \pi_{\text{rt}}, \mathbf{w}\rangle$ to the responsible agent. Safety mechanisms are procedural: exploration is canaried, actions are envelope-clipped, reroutes are shadow-checked, harmful actions trigger rollback, and p is tracked as a moving average of prediction accuracy.

F. Analysis Scope and Assumptions

The results below are conditional and operational; they characterize predictable behavior, not global optimality. We assume: (A1) within each exploration window, each visited state region has stable expected utility with bounded measurement noise; (A2) the stream visits a finite set S of regions often enough for the cache to receive repeated samples; (A3) canary and micro-exploration continue throughout; (A4) agents enforce the latest envelope and the controller serializes conflicting meter-rate updates.

Theorem 1 (Single-Agent Cache Consistency)**.** *Under A1–A3, after collecting $N_{\text{conv}} = O\left(\frac{R^2}{\varepsilon^2}\ln\frac{|S|}{\delta}\right)$ samples, the HAT predicts $a^\dagger(\mathbf{s})$ for all visited regions with probability at least $1-\delta-\eta$ (η bounds exploration-noise mislabeling). Once accuracy exceeds p_{th} by a margin, the agent switches to model execution after $O(1/(1-\beta))$ additional samples.*

Theorem 2 (Policy-Envelope Compliance)**.** *Every action executed by an EnvelopeTC agent satisfies $\langle r_{\min}, r_{\max}, \pi_{\text{reroute}}\rangle$, provided the agent has received the envelope and applies the enforcer before OVSDB execution. This is a syntactic guarantee and does not imply SLA or network-wide invariants under stale telemetry or conflicting reroutes.*

Theorem 3 (Conditional Recovery after Drift)**.** *If path conditions change at t_d and restabilize, and the HAT error rate rises from p_1 to $p_2 > p_1$, ADWIN detects the drift within $T_{\text{detect}} = O\big((p_2-p_1)^{-2}\big)$ samples (w.h.p.). The agent then recovers cache consistency after N_{conv} additional labeled samples under A1–A3.*

V. Implementation

EnvelopeTC is implemented as three software components.

SDN Controller (Ryu + EnvelopeTC Module). We extend Ryu [8] with an EnvelopeTC controller module for global monitoring, policy envelope generation, bottleneck detection, and multi-agent coordination. The module subscribes to sFlow records from OVS instances and maintains per-path envelope state in an in-memory store.

Edge Agent (Python Daemon). Each agent runs as a Python daemon co-located with the vSwitch control plane. It provides: (i) a telemetry collector polling OVS port counters and queue statistics via OVSDB every 50 ms; (ii) a HAT model per monitored path using the River ML library [9]; (iii) an action executor translating model outputs to meter/queue/flow-table updates via OVSDB RPC; and (iv) a policy-envelope enforcer clipping all actions to controller-provided bounds. One-way delay is measured from timestamped endpoint probes, not inferred from OVSDB counters.

Controller–Agent Transport (gRPC). Policy envelopes and reports are exchanged over gRPC. Envelopes are pushed on change; agents apply updates within a 500 ms staleness bound.

Action dimensions. The prototype implements three action dimensions: (1) OpenFlow meter rate adjustment ($\alpha = 10\%$, step size), (2) OVS queue assignment across three priority levels, and (3) elephant-flow rerouting via priority flow-table rules.

VI. Evaluation

We evaluate EnvelopeTC on overall performance, tail latency, overhead, and robustness under controller and telemetry stress.

A. Methodology and Baselines

Experiments run on 1,024 AWS c5.xlarge instances emulating a 64-rack Clos fabric with OVS bridges, GRE tunnels, 10 Gbps `tc`-shaped logical links, and OpenFlow group-table routing. A Ryu controller on a c5.2xlarge manages 64 rack agents over gRPC; agents poll OVS counters every 50 ms and enforce a 500 ms reroute cooldown. Long flows use iperf3; short flows use fixed-seed replay across rack pairs. We report 10 trials (seeds 1000–1009) with 95% Student-t confidence

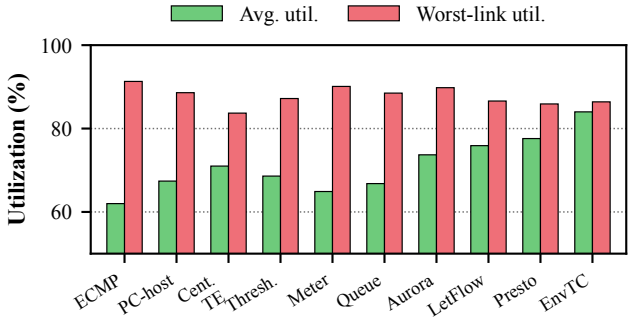
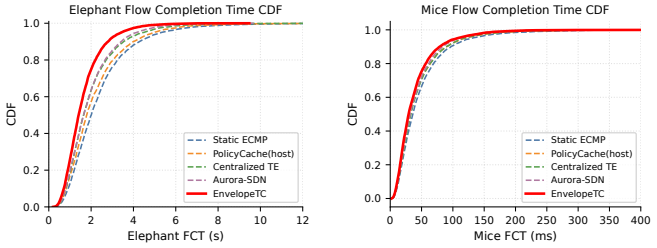


Fig. 2. Link utilization: average vs. worst-link maximum across all schemes. EnvelopeTC achieves the highest average utilization with a small max-average gap.



(a) Elephant FCT (>10 MB) (b) Mice FCT (<100 KB)

Fig. 3. Flow Completion Time CDFs (elephant-heavy workload). EnvelopeTC shifts the elephant CDF leftward by 33% vs. Static ECMP while also improving mice FCT.

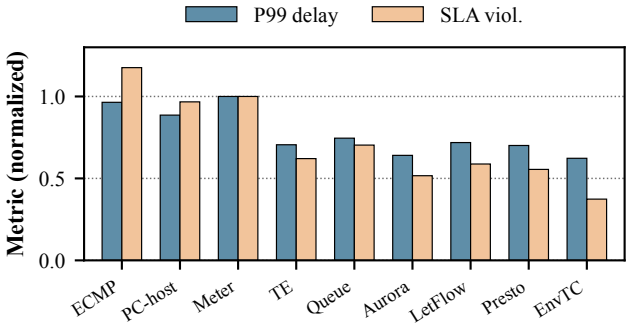


Fig. 4. Tail latency and SLA compliance under the mixed real-time workload, normalized to Static Meter. EnvelopeTC has both the lowest P99 delay and the lowest violation rate.

intervals; FCT comes from application logs and one-way delay from 100 Hz timestamped probes. Because this is a software public-cloud fabric, absolute latency and reordering include VM, GRE, and OVSDB noise and should be read as relative comparisons.

All schemes share the same topology, traces, and measurement pipeline. Baselines are Static ECMP, Centralized TE with a 500 ms loop, Threshold Reroute above 80% utilization, Static Meter, Heuristic Queue, frozen Aurora-SDN [4], LetFlow-style flowlets [10], Presto-style flowcells [11], and PolicyCache end-host congestion control [6]. Aurora-SDN is trained offline for 10^6 steps and evaluated without online fine-tuning; claims against learned baselines are limited to this frozen setting and to PolicyCache's end-host learner.

We use three workloads: **elephant-heavy** (80% of bytes from flows >100 MB; matrix shifts every 60 s), **mice-heavy** (Pareto mean 50 KB, κ=1.2), and **mixed real-time** (40% latency-sensitive flows with a 10 ms one-way SLA).

B. Link Utilization

Figure 2 shows average and worst-link utilization under the elephant-heavy workload.

EnvelopeTC achieves **35.5%** higher average utilization than Static ECMP (84.0% vs. 62.0%) and **18.3%** higher than Centralized TE. Its worst-link maximum is 86.4%, only 2.4 pp above its average. PolicyCache (end-host) yields only 67.4%: end-host CC cannot see bottleneck locations or reroute elephant flows, confirming that network-level intra-path learning provides capabilities beyond end-host control. Aurora-SDN performs well within its training distribution but degrades by 6.8 pp under shifted traffic matrices. LetFlow-style and Presto-style improve over Static ECMP but remain below EnvelopeTC under sustained hot spots.

C. Flow Completion Time

EnvelopeTC reduces elephant mean FCT by **33.2%** and P99 FCT by **40.3%** relative to Static ECMP. The improvement over Centralized TE (18.7% mean, 24.1% P99) stems from faster reaction: edge agents detect queue buildup within the current 50 ms interval and execute reroutes within the same RTT, whereas Centralized TE requires a full sFlow aggregation cycle (≈500 ms). PolicyCache (end-host) reduces elephant mean FCT by only 9.5% relative to Static ECMP because it cannot reroute elephant flows, a fundamental limitation of end-host CC.

D. Tail Latency and SLA Compliance

EnvelopeTC reduces P99 delay for real-time flows by **37.7%** compared to Static Meter and by 11.7% compared to Centralized TE. SLA violation rate drops from 18.2% (Static Meter) to **6.8%**, a 62.6% relative reduction. PolicyCache (end-host) achieves 17.6% violation rate; it reduces P99 delay slightly by backing off cwnd but cannot promote flows to higher-priority queues, a capability unique to SDN-level control. Aurora-SDN achieves 9.4% within its training distribution but rises to 13.1% under out-of-distribution traffic, consistent with the limitations of frozen offline policies in this setup.

E. Convergence, Overhead, and Coordination

All agents reach model execution mode within 400 ms from cold-start (Figure 5). Agent CPU stays below 2.1% and memory below 13.4 MB. Without controller arbitration, agents sharing a bottleneck oscillate at 4.3 reroute-flip events/s with ±22 pp utilization variance; with arbitration, flips drop to 0.4/s and

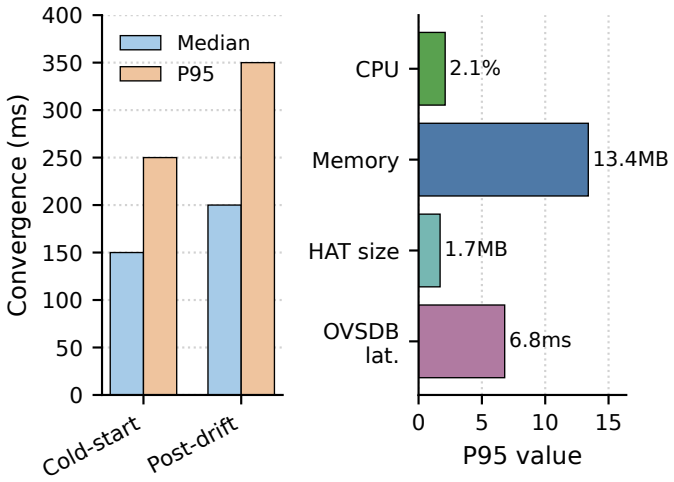


Fig. 5. Agent convergence and overhead summary. Convergence remains below 350 ms at P95, with small per-agent CPU, memory, model-size, and OVSDB-latency overheads.

variance narrows to ± 5.1 pp, adding $<0.3\%$ to controller–agent traffic.

F. Ablation and Sensitivity

We ablate each action type on the elephant-heavy workload. Rerouting is most impactful: disabling it reduces average utilization from 84.0% to 76.3% (-7.7 pp) and elephant mean FCT from 1.61 s to 2.09 s. Queue-priority contributes $+3.9$ pp and metering $+1.9$ pp. EnvelopeTC is robust across parameter settings: at α=10% (default), utilization gain over Static Meter is 19.1 pp; α=2% reduces it to 6.1 pp, and α=25% speeds convergence but adds 2.4 pp SLA violations. Coarsening the interval from 50 ms to 200 ms costs 4.7% in utilization and 8.2% in P99 FCT. Under shifted traffic matrices, ADWIN-triggered re-exploration converges within 200–350 ms.

G. Controller Loop and Robustness

Centralized TE update interval. TE-loop sensitivity shows that faster loops improve utilization but increase CPU and rule churn sharply. Even a 50 ms loop remains below EnvelopeTC: global collect–optimize–install cannot match our design's split 50 ms agent / 500 ms envelope cadence.

Stability and side effects. Stress tests cover stale envelopes, telemetry loss, slow OVSDB commits, controller outage, rapid traffic oscillation, and excessive reroute frequency. During a controller outage, agents continue meter and queue actions inside the last valid envelope but disable new reroutes after the 500 ms staleness bound. Disabling the reroute cooldown improves neither utilization nor FCT but increases flips and reordering.

H. Discussion and Limitations

The results suggest that EnvelopeTC is most useful when the controller can express operator intent as narrow action bounds and local agents can observe enough short-term telemetry to choose among those legal actions. This matches software fabrics, virtualized edge deployments, and SmartNIC-assisted designs where meter, queue, and selective reroute actions are already available to the control plane. The design is less appropriate when the data plane exposes only static ECMP weights or when telemetry arrives slower than the congestion episode being controlled.

In deployment, envelope width is the main operator knob. Narrow envelopes reduce risk and make rollback easier, while wider envelopes allow faster recovery from traffic-matrix shifts but require stricter audit and arbitration. This separation lets operators tune safety policy at the controller without retraining the edge learner.

Our evaluation intentionally separates relative system behavior from hardware claims. The OVS/GRE testbed exercises controller-agent timing, online adaptation, reroute coordination, and failure handling at scale, but it does not reproduce ASIC queue dynamics, lossless RDMA fabrics, or in-switch congestion feedback. We therefore view hardware P4 or SmartNIC validation as a deployment step rather than as evidence needed for the core control abstraction. A second limitation is baseline scope: Aurora-SDN is a frozen offline-RL baseline, so the comparison tests robustness to distribution shift, not the full design space of online-adapting multi-agent RL.

VII. Related Work

SDN traffic engineering and load balancing. B4 [12], SWAN [13], and Hedera [14] optimize routing at controller timescale, while CONGA [15], HULA [16], LetFlow [10], and Presto [11] address ECMP imbalance through in-network feedback, flowlets, or flowcells. EnvelopeTC is complementary: it keeps global policy at the controller but gives edge agents bounded authority over SDN meters, queues, and selective reroutes.

Edge admission and defense. Edge-side mechanisms can also adapt ingress behavior for security: EdgePoW [17] uses ingress-aware non-interactive proof-of-work to defend against volumetric SYN floods. EnvelopeTC targets a complementary control problem, using bounded edge authority for adaptive traffic control rather than attack admission.

Online learning and hierarchical control. Vivace [18], Proteus [19], and PolicyCache [6] show that online exploration can adapt transport behavior inside a local scope. EnvelopeTC lifts that locality to SDN path aggregates, where actions can affect tenants and neighboring paths; policy envelopes and arbitration make such local learning auditable. ONIX [20] and Kandoo [21] similarly layer control for scalability, but their local elements process events rather than execute bounded online learning policies.

VIII. Conclusion

We presented EnvelopeTC, a hierarchical SDN traffic-control framework that lifts intra-flow locality from end-host congestion control to SDN traffic aggregates. The main systems contribution is not a new tree learner, but the controller–agent abstraction needed to make local online learning composable in SDN: policy envelopes bound exploration and execution, action logs make local decisions auditable, and controller arbitration resolves conflicts on shared bottlenecks.

Evaluation on a 1,024-host cloud testbed shows 35.5% higher average core-link utilization than Static ECMP, 40.3% lower P99 elephant FCT, and 62.6% fewer SLA violations compared to Static Meter, with under 2.1% per-agent CPU overhead and 12 MB memory footprint. Multi-agent coordination eliminates reroute oscillation with negligible additional control traffic.

The main lesson is that locality alone is not enough: local learners become deployable in SDN only when their authority is explicitly bounded, versioned, and auditable. Future work includes hardware-fabric validation with P4 or SmartNIC feedback, stronger online-adapting MARL and online-RL baselines, extensions to multi-domain and inter-AS traffic engineering, and stronger multi-agent analysis for discrete rerouting and queue-priority actions.

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